REVIEW OPEN ACCESS

Intelligent Maintenance Approaches for Improving Photovoltaic System Performance and Reliability

Demetris Marangis[1,2] | Georgios Tziolis[1,2] | Andreas Livera[1,2] | George Makrides[1,2] | Andreas Kyprianou[1,2] | George E. Georghiou[1,2]

[1]PV Technology Laboratory, Department of Electrical and Computer Engineering, University of Cyprus, Nicosia, Cyprus | [2]PHAETHON Centre of Excellence for Intelligent, Efficient and Sustainable Energy Solutions, Nicosia, Cyprus

Correspondence: Demetris Marangis (marangis.demetris@ucy.ac.cy)

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ABSTRACT

Photovoltaic (PV) systems play a pivotal role in the transition to renewable energy worldwide, yet their long-term performance and cost-effectiveness critically depend on robust Operation and Maintenance (O&M) strategies. While corrective and preventive maintenance have seen significant progress, the development of predictive analytics that proactively generate warnings to anticipate underperformance issues and potential failures remains underexplored. This article makes a substantial contribution by providing a comprehensive review of maintenance approaches, including corrective, preventive, predictive, and extraordinary, with a special focus on the integration of predictive analytics for smart maintenance in PV systems. The study evaluates how cutting-edge technologies, such as the Internet of Things (IoT) and Artificial Intelligence (AI), facilitate real-time monitoring, diagnostics, and automated early warning systems to anticipate underperformance issues and potential failures, thereby enabling proactive maintenance scheduling. By summarizing the capabilities of these intelligent monitoring systems, the article demonstrates how predictive analytics can significantly reduce unexpected downtime, enhance decision-making, and ultimately lower the levelized cost of energy (LCOE) of PV assets. Finally, the article provides recommendations and outlines future directions for the development of standardized frameworks to optimize smart maintenance practices and improve solar asset management, advancing the state-of-the-art in predictive analytics for the PV industry.

1 | Introduction

The transition to sustainable energy is crucial to address climate change, with solar photovoltaics (PV) emerging as a key clean energy technology due to their ease of installation and the declining PV module costs [1]. Solar PV technology is widely adopted in commercial, industrial, and residential sectors and can be rapidly deployed to meet global future energy needs [2]. The installed PV capacity across the European Union (EU) has grown from 11.3 GW in 2009 [3] to 1 TW in April 2022 [4], recognizing the enormous solar potential for independent energy supply, climate protection and alleviation of pressure on energy prices. It was doubled to 2 TW by November 2024, and it is projected to exceed 5 TW by 2028 [5].

Despite considerable growth and advancements in PV technology, solar systems remain vulnerable to defects from environmental, structural, electrical, equipment and other factors. Power losses caused by extreme weather conditions and equipment malfunctions have risen to 94% since 2019 [6], with further increases expected as PV modules age. Without effective Operation and Maintenance (O&M), the global PV industry could lose $14.5 billion in 2024 [7]. Early detection of underperformance is crucial for long-term efficiency [8], and O&M

practices can recover about 5% of lost energy [9], translating to significant financial gains [10]. However, a recent study [11] shows that PV systems still underperform by about 8%, highlighting the ongoing need for robust O&M strategies.

Building on the importance of effective O&M for mitigating PV system underperformance and financial losses, maintenance strategies are generally divided into four categories (see Figure 1): preventive, corrective, predictive, and extraordinary maintenance [7]. These strategies are designed to keep systems operating optimally, minimize the duration of faults and downtimes, and ultimately reduce energy production losses [7]. More details for the different maintenance types are provided in Section 2.

While traditional O&M strategies (preventive, corrective, and extraordinary maintenance) are essential for minimizing PV system downtime and energy losses, they often rely on scheduled or reactive interventions. To further enhance reliability and efficiency, the PV industry is now turning to prognostic maintenance, utilizing advanced predictive and prescriptive analytics [12].

These modern analytic techniques enable early detection and prediction of faults while also providing recommendations and decision actions (based on forecasts) to prevent equipment failures and to mitigate potential risks of PV underperformance, paving the way for smarter and more proactive maintenance. By integrating the aforementioned analytics to intelligent monitoring systems (see Figure 2), PV operators can transition from conventional maintenance practices to predictive and prescriptive analytics, unlocking new opportunities for optimizing performance and reducing operational costs by taking proactive and corrective measures in a time- and condition-based manner.

Despite significant advancements in PV system maintenance strategies, the integration of emerging technologies like artificial intelligence (AI) analytics and the Internet of Things (IoT) remains underexplored in current literature, especially in the field of predictive and prescriptive analytics [13–15]. In this domain, only a handful of studies [16–18] have addressed the application of AI-IoT for predictive maintenance in PV systems. For example, Onwusinkwue et al. [16] reviewed AI techniques such as deep learning and neural networks for predictive maintenance, discussing both the challenges and opportunities associated with AI implementation in renewable energy systems. Likewise, Ledmaoui et al. [17] highlighted recent progress in predictive maintenance methods for PV systems, emphasizing the role of incorporating IoT monitoring and Machine Learning (ML) techniques to improve the accuracy and effectiveness of maintenance procedures. Finally, a comprehensive review of ML forecasting methods to support predictive maintenance in PV systems was presented in [18]. This research gap is critical, as intelligent AI-IoT monitoring systems that leverage diagnostic, predictive, and prescriptive analytics have the potential to revolutionize O&M by enabling earlier fault detection, optimized maintenance scheduling, and reduced operational costs. This review article thus addresses the pressing need to move beyond conventional O&M by analyzing the synergy between AI and IoT for smart maintenance in PV systems. Specifically, it examines how integrating AI-driven analytics with IoT-enabled monitoring

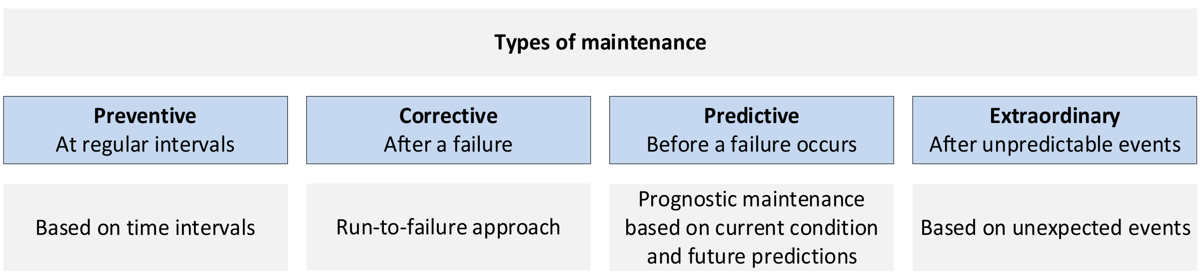

FIGURE 1 | Maintenance types.

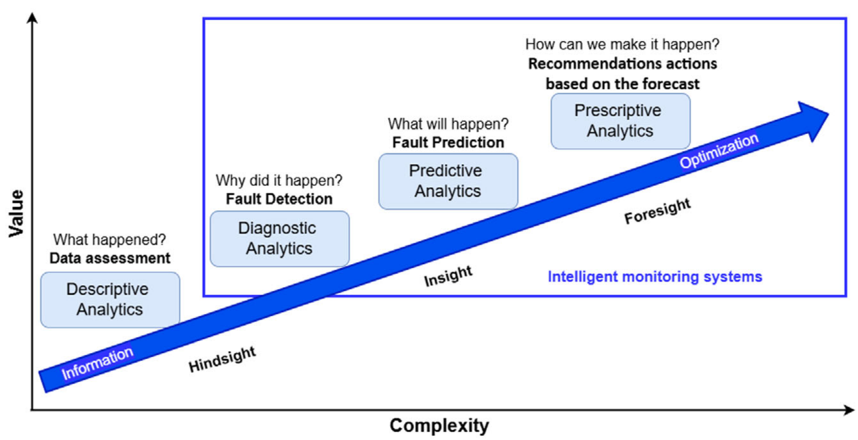

FIGURE 2 | Value and complexity of analytic techniques. It shows the analytic progression moving from information to optimization in intelligent monitoring systems.

transforms solar maintenance from traditional reactive approach to an intelligent, predictive one. By synthesizing the latest research and industry practices, this article provides a comprehensive framework for implementing smart maintenance strategies that enhance PV performance, minimize maintenance downtimes, and maximize asset lifespan. The study's findings offer actionable recommendations for PV operators, technicians, and researchers seeking to adopt AI-IoT solutions to improve solar asset management, while also enabling proactive actions in PV systems.

The rest of this article is structured as follows: Section 2 provides the search strategy and an overview of the reported corrective, preventive, predictive, and extraordinary maintenance strategies for PV systems. It mainly focuses on smart maintenance strategies and their capabilities. Section 3 discusses the main findings of the present study and provides recommendations and future perspectives to improve solar asset management through the formulation of a comprehensive framework for smart maintenance strategies. Focus is shed mainly on intelligent PV predictive maintenance. Finally, Section 4 concludes this review study.

2 | Intelligent Maintenance Approaches

2.1 | Search Strategy and Current State of Literature

To quantify the current literature state, the Scopus, IEEE Xplore, Google Scholar, and EUPVSEC databases were used to assess the volume of publications available for each maintenance category (corrective, preventive, predictive, and extraordinary) that incorporate intelligent methodologies based on AI analytics. The advanced search queries used for each maintenance type are detailed in Table 1.

This review was conducted in accordance with the Preferred Reported Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines [12], as shown in Figure 3, which includes the four following phases: identification, screening, eligibility, and inclusion. Specifically, a total of 1651 articles were used in the bibliometric analysis and screened, out of which 953 were excluded due to duplication, non-English language, being irrelevant, editorial, erratum, or retracted articles. The remaining 698 articles were checked for eligibility criteria, where 544 articles were excluded due to their focus on control strategies, modeling and simulation, data fusion, feature selection, etc. Finally, 154 articles were included in the systematic review, comprising 93% journal articles and conference publications, 5% review articles, and 2% books.

Of the articles included in the review, most of them (93%) addressed corrective maintenance strategies, while significantly fewer focused on predictive maintenance (4%) and preventive maintenance (3%). These findings highlight a notable gap in research on proactive maintenance approaches, which could substantially reduce downtimes, facilitate early identification and prediction of underperformance issues, and ultimately enhance overall system performance.

The number of articles meeting the initial eligibility criteria (total 698) and focusing on intelligent maintenance has skyrocketed in recent years (see Figure 4). In 2024 alone, 168 articles addressed intelligent maintenance, a dramatic increase from just 6 articles in 2016. This surge reflects rapid advancements in AI applications and the growing availability of data, both of which have enabled researchers to leverage advanced techniques for deeper insights into the system's health and operational status. This trend highlights the increasing importance and adoption of smart solutions for refining O&M strategies in the PV sector. These advancements empower solar industry stakeholders to make informed decisions, enhance operational efficiency, and drive sustainable growth within the renewable energy landscape. As these technologies become more widespread, integrating AI and IoT into intelligent monitoring systems is now recognized as essential for improving PV system reliability and performance. By harnessing real-time data and advanced analytics, the combination of AI-IoT offers significant opportunities to improve decision-making, boost operational efficiency, extend the lifespan of PV assets, and optimize maintenance scheduling.

2.2 | Preventive Maintenance

Preventive maintenance involves regular visual and physical inspections, functional testing, measurements, and verification

TABLE 1 | Advanced search queries used in databases for each maintenance type. The "TITLE-ABS-KEY" operator in the databases allows for a comprehensive search by identifying articles where the specified terms in the parentheses appear in the title, abstract, or keywords.

Type of maintenance	Advanced search (TITLE-ABS-KEY)
Preventive	("preventive maintenance") AND ("photovoltaic" OR "PV") AND ("intelligent" OR "machine learning" OR "artificial intelligence" OR "AI")
Corrective	("corrective maintenance" OR "fault detection" OR "anomaly detection" OR "failure detection" OR "fault classification" OR "failure classification") AND ("photovoltaic") AND ("intelligent" OR "machine learning" OR "artificial intelligence" OR "AI")
Predictive	("predictive maintenance" OR "predictive analytics" OR "fault prediction" OR "failure prediction") AND ("photovoltaic") AND ("intelligent" OR "machine learning" OR "artificial intelligence" OR "AI")
Extraordinary	("Extraordinary maintenance") AND ("photovoltaic" OR "PV") AND ("intelligent" OR "machine learning" OR "artificial intelligence" OR "AI")

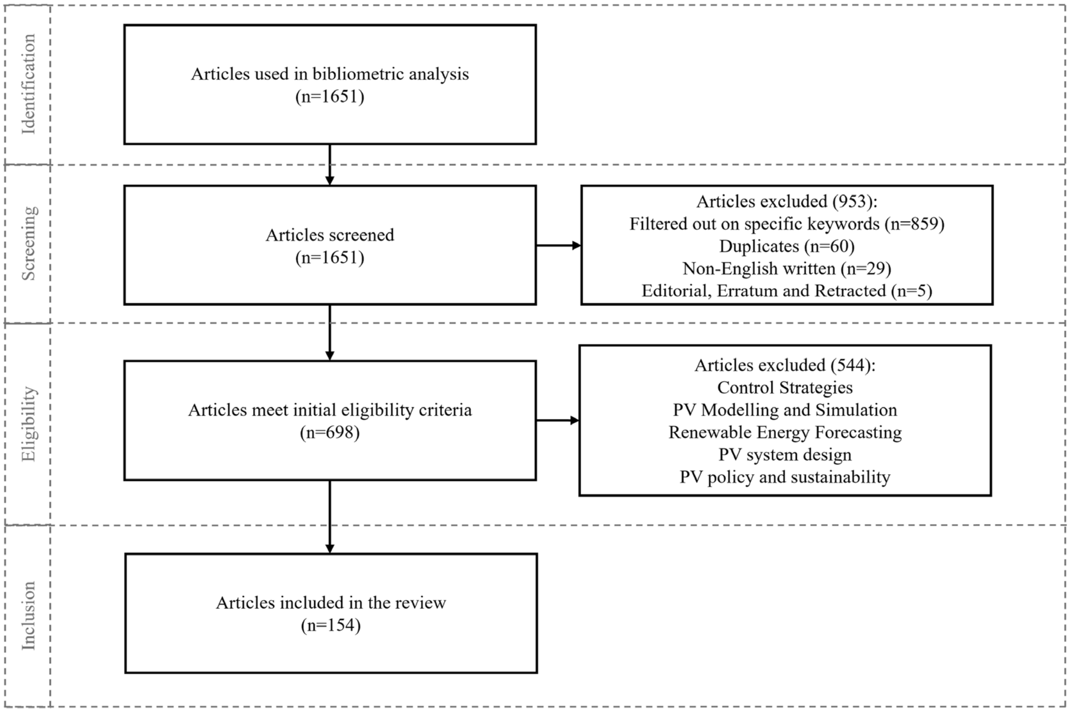

FIGURE 3 | Methodology based on PRISMA guidelines, illustrating the selection process of studies in the review article.

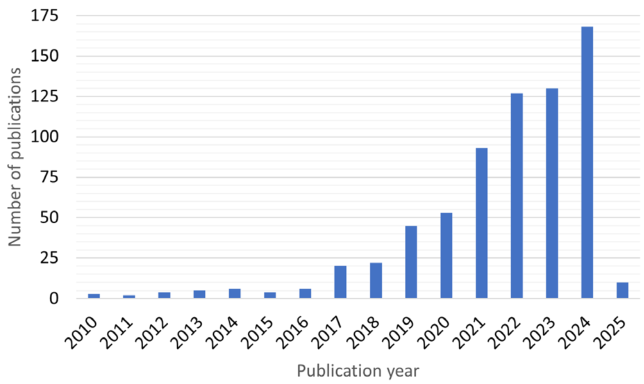

FIGURE 4 | Number of PV intelligent maintenance articles published between 2010 and 2025.

processes to confirm adherence to operating guidelines and warranty conditions [19–22]. This type of maintenance is carried out at predetermined regular intervals according to Original Equipment Manufacturers (OEM) manuals and O&M plans, aiming to prevent failures and maintain the health of system components at a desired service level. Typical examples include routine inspections (such as cleaning PV modules) at set timeframes following the operation manuals [23], thermal inspections to identify PV module failures (e.g., hot spots, bypass diode, etc. [24]), and power losses, on-site measurements (e.g., string measurements and current–voltage - *I–V* curve tracing) as well as ad hoc replacement of and calibration of inverters or sensors [25].

These activities enable PV operators to monitor the system's operational status and implement necessary measures that slow down the natural degradation of PV modules. Consequently, preventive maintenance not only extends the lifecycle of assets but also reduces system downtime. Moreover, in certain cases, preventive maintenance can adopt a proactive approach, for instance, applying antisoiling coatings before system commissioning [26, 27] or optimizing module performance and system design to minimize dust accumulation. Such proactive measures are typically implemented before any failure occurs, aiming to prevent issues from arising altogether.

In recent years, the focus of preventive maintenance has shifted toward optimizing inspection intervals to reduce maintenance costs while improving system's energy yield [20]. For example, regular monthly cleanings or cleanings triggered by soiling thresholds have been shown to reduce annual energy losses caused by soiling from as high as 30% down to approximately 4% [28]. Building on this, preventive maintenance can be further refined by incorporating reliability data and failure probabilities. This approach allows technicians to perform maintenance actions only when certain risk thresholds are met, thus optimizing cost and resource use. Sa'ad et al. [29] exemplified this concept by developing a MATLAB-based selective preventive maintenance algorithm based on reliability thresholds to determine optimal maintenance schedules, aiming to maximize system availability while minimizing costs. The results showed the optimal number of preventive maintenance actions to be 5 and a maintenance cost of 1649 monetary units with a maximum availability of 99.75%. Another study [30] also used reliability thresholds to optimize the preventive maintenance interval of PV equipment. The optimal maintenance strategy (i.e., repairing the equipment 4 times at an incomplete maintenance threshold of 0.8) decreased the costs by up to 20.3% compared to average maintenance expenses and increased availability by up to 0.2395%.

Despite these advancements, the overall costs of preventive maintenance remain relatively high, primarily due to the extensive number of site inspections required, especially in large-scale PV installations [31, 32]. In response, the industry is increasingly turning to advanced diagnostic tools and automation, such as thermographic inspections and autonomous portable robots to streamline maintenance and reduce expenses. Thermographic inspections, for example, allow maintenance teams to map temperature anomalies across each PV module. This enables the early identification of minor electrical resistances or material degradations before they escalate into catastrophic failures or fire hazards. When integrated with AI-driven image-processing software, thermography can be automated and deployed via drones, facilitating scalable inspections that minimize downtime, lower operational costs, and maximize energy yield [24]. Simultaneously, the integration of robotics is revolutionizing PV maintenance practices. Recent advances, such as the development of portable PV cleaning robots [33] demonstrate how smart robotics can dramatically reduce the time and labor required to maintain large PV parks. Instead of dispatching crews across vast arrays, self-powered robots autonomously navigate the solar park using intelligent control and path planning, recharging from the sun as they work. Their adjustable cleaning heads adapt to each panel's angle for thorough coverage, while remote-monitoring interfaces enable a single operator to oversee multiple machines at once. This transition from labor-intensive manual cleaning to agile robotic solutions not only increases efficiency but also makes frequent cleaning more cost-effective, thereby enhancing overall system performance.

Supporting these technological shifts, Jiang et al. [34] recommended the adoption of an automated and low-cost solution to replace regular manual cleanings for large-scale PV installations. As such, an intelligent O&M cleaning robot was developed and used along with a cleaning cycle optimization method. The optimization results showed that using the robot to clean PV modules with dust accumulation of 4, 6 and 8 g/m^2 yielded cleaning efficiencies of 92.14%, 91.04%, and 90.09%, respectively. Compared with manual cleaning, the optimal cleaning cycle was reduced from 48.3 days to 26.5 days when using the robot, while the annual cleaning costs were lowered by approximately 84% for a 2 MW PV plant. Collectively, these findings strongly support the implementation of intelligent, automated, and robot-based solutions to improve the economic benefits of utility-scale PV plants.

2.3 | Corrective Maintenance

Corrective maintenance encompasses all activities aimed at restoring a PV system following a failure event or repairing malfunctioning components and damages [35]. This type of maintenance is typically initiated after a fault is detected, either through remote monitoring systems or during routine site inspections, with the primary goal of minimizing revenue loss caused by system downtime [1, 35–38]. This is achieved by reducing fault response and resolution (repair) times (see Figure 5).

Since corrective actions are unplanned and often urgent, they tend to be associated with higher costs due to both lost power generation and potential damage to system components. Nevertheless, these actions are essential for addressing detected issues (such as faulty inverters or degraded PV modules) and restoring the PV system back to normal operation.

Corrective maintenance generally involves fault diagnosis as well as temporary or full repairs. It can be divided into three levels of intervention: (i) intervention without the need for component substitution, (ii) intervention requiring component replacement, and (iii) intervention involving software adjustments.

A critical aspect of corrective maintenance is the effective application of fault diagnosis methods. Various techniques are used to support operators in this process, primarily by facilitating fault identification, forecasting production losses, and monitoring the overall health of PV systems.

These techniques can be broadly categorized into visual inspection, electrical data analysis, and image processing [39]. Among these, visual inspection is the simplest method and the most straightforward approach for detecting visible defects (e.g., delamination [40]) and failures within PV installations. However, visual inspection is limited to identifying only visible issues such as physical damage, dirt accumulation, and shading and cannot detect more subtle faults like potential-nduced degradation (PID) [41]. Additionally, frequent visual inspection checks can be time-consuming, especially for large-scale PV plants.

Given the rapid growth in the number and capacity of PV systems, there is an increasing need for intelligent methods to enable timely and accurate fault diagnosis. In this domain, advanced techniques, such as anomaly detection, have been developed to assess system's health and to identify faults in real-time [42, 43]. The insights and fault information generated by these methods empower PV plant operators to streamline O&M activities (i.e., corrective actions), ultimately enhancing system reliability and performance [44–47]. These methods primarily build upon two diagnostic approaches: electrical analysis and image processing.

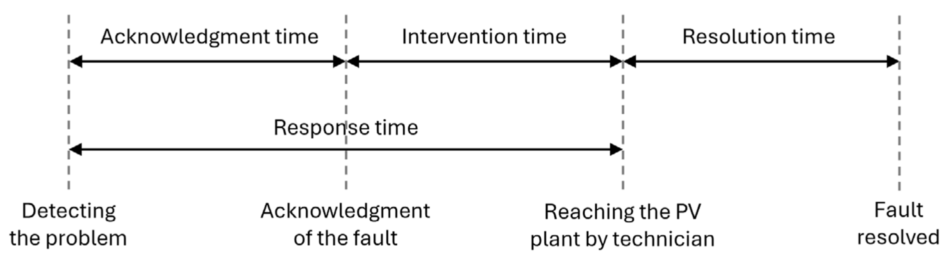

FIGURE 5 | Response and resolution times.

Electrical analysis methods rely on numerical, electrical signature and statistical analyses to detect PV failures. Conversely, image processing techniques rely mainly on the analysis of thermal and electroluminescence (EL) images [39, 48–50]. Recent advances in data science and computer vision have revolutionized both approaches, with AI algorithms now enabling the development of fast and accurate data- and image-driven fault detection algorithms [51–55]. In data-driven diagnostics, AI-based techniques are used to simulate *I–V* and power-voltage (*P–V*) curves [47], predict the PV performance, and identify root causes of faults. These algorithms have demonstrated exceptional accuracy in classifying various faults and underperformance issues, such as partial shading [56], soiling [57], and module- and inverter-related failures [58]. Underperformance conditions are commonly identified by analyzing the deviations between the actual and simulated PV performance [59, 60], using either statistical threshold levels [59, 60] or outlier detection techniques to identify data anomalies [61, 62]. For image-driven diagnostics, infrared (IR), EL, photoluminescence (PL), and ultraviolet (UV) fluorescence images of operating PV modules are typically captured and analyzed by advanced ML algorithms [38, 63–65]. Processing these IR, EL, PL, and UV images enables the geolocation and identification of hot spots, inactive PV cells due to bypass diode failures, cracks, shunts, and other defects, while also diagnosing degradation modes such as PID and light-induced degradation (LID) with high fault detection accuracies [53, 54, 66].

Finally, the electrical- and/or image-based fault diagnostic methods are incorporated into decision support systems (DSS) that generate recommendations for field corrective actions based on the detected underperformance incidents [7, 67, 68].

2.3.1 | Electrical Data Analysis

It involves the analysis of key electrical parameters (such as voltage, current, and power output) and characteristic curves, in parallel with weather data and other performance metrics. This comprehensive analysis enables the identification of fault issues and the implementation of targeted corrective actions.

The use of simulation, electrical circuit models (e.g., single-diode model), and characteristic curves (*I–V* and *P–V*) provides valuable insights about the PV health status, also enabling the identification of different faults [69]. For example, Voutsinas et al. [52] used the single-diode model with five parameters along with ML principles for the early detection of mismatch, open- and short-circuit faults in PV systems, reaching an accuracy of 97.11%. Additionally, Hocine et al. [70] used the *P–V* and *I–V* characteristics to detect different degradation modes such as browning of PV cells.

Furthermore, monitoring key PV electrical parameters enables the detection of various PV module and inverter related faults. These include partial shading, bypassed PV modules, open- and short-circuit faults, inverter disconnection, and other PV underperformance issues. As such, models that analyze deviations between measured and predicted/estimated electrical performance were developed for identifying malfunctions within the PV system. In [71], a procedure was presented to detect open- and short-circuit conditions using a statistical comparative assessment between measured and simulated DC power. The current and voltage indicators (ratio between measured and simulated quantity) were used to classify the detected faults. For fault identification, the use of statistical thresholds such as the 1 standard deviation (σ) proved effective in diagnosing short-circuit and open-circuit fault conditions in PV systems [71]. A comparative analysis of the voltage, current, and power output under simulated and measured conditions successfully identified short- and open-circuit conditions with accuracies >98%. Moreover, the use of probability modeling and confidence intervals has also been effectively used to detect various types of faults such as short-circuit and partial shading with accuracies >97% [55].

Due to the rise of AI, an increasing number of data-driven fault diagnostic algorithms have been developed [1, 47, 72, 73]. Livera et al. [47] evaluated the performance of different AI algorithms (decision trees—DT, k-nearest neighbors—k-NN, support vector machine—SVM, and fuzzy inference systems—FIS) for detecting open and short-circuit faults, partial shading, inverter shutdown, and bypass diode failures. The developed algorithms achieved high detection and classification accuracies. Khalid et al. [1] used regression analysis and a random forest classifier to detect partial shading, open- and short-circuit faults. In another approach, the extreme learning machine (ELM) method was used for detecting malfunctions in the grid network, inverter faults, sensor, and MPPT controller errors, achieving an average accuracy of 77.83% [72]. In another instance, an ANN model was developed to model the soiling rate [74], while the SVM algorithm was used to detect soiling conditions, achieving an accuracy of 80% [75].

Despite their high accuracy, the AI data-driven fault diagnostics require labeled and a large amount of training data, but such data is often scarce in real-world PV applications. To address this limitation, synthetic data, which mimic both faulty and healthy system behavior, can be used for training and classification purposes [76]. For instance, a conventional neural network (CNN) algorithm trained on combined field and synthetic data demonstrated increased accuracy and robustness for separating healthy and faulty panels [77]. This approach not only replicated real fault behaviors but also outperformed purely data-driven models under limited data availability. Likewise, Sarquis Filho et al. [76] demonstrated that even with synthetic datasets, the trained neural network model could achieve high accuracies (>90%) for diagnosing short-circuit and bypass diode short-circuit faults.

Overall, the AI data-driven models require clean data and consistent updates to remain effective and accurate. The performance of AI-driven systems heavily depends on the quantity, quality, and diversity of the available data. AI systems heavily rely on large volumes of high-quality data for the training, validation, and testing. Moreover, AI benefits from diverse datasets to ensure robustness and generalization across different fault conditions. The available data are thus used to train AI models to recognize fault patterns and to make accurate PV performance predictions. Such models continuously improve their accuracy through iterative learning processes. As more operational data become available, the models refine their predictive capabilities, reducing false positives and enhancing the corrective maintenance scheduling precision. Without sufficient and high-quality data, AI models may fail to generalize well to different fault situations, leading to poor performance and unreliable outcomes for fault detection and classification. Finally, data preprocessing techniques [78, 79] are essential to clean, normalize, and prepare the data for the AI models, while labeled data are crucial for supervised learning tasks, facilitating accurate model predictions for fault detection and classification.

2.3.2 | Image Processing Algorithms

Image processing techniques serve as a powerful alternative for detecting and locating PV faults, such as failed interconnection, resistive soldering bonds, cracks, damaged/inactive PV cells, bypass diode failures, hot spots, and degradation mechanisms [80]. The efficacy of imaging techniques for inspection and corrective maintenance is well established [81], with EL and thermal imaging playing a particularly critical role. Early detection of high-temperature regions through thermal imaging or dark to black areas in EL imaging is essential to identify faults and problems present or developing within the PV module that directly impact system reliability and longevity [82, 83].

Researchers have already used EL and IR imaging with AI for fault detection and classification. In [84], the CNN network was utilized with t-distributed stochastic neighbor embedding (t-SNE) and Uniform Manifold Approximation and Projection (UMAP) to classify the IR images of faulty and healthy PV modules. The algorithm was capable of detecting junction box failures, patchwork, substring, multistring, and hot spots, reaching an accuracy of 99.81%. The use of transfer learning technique [85] enabled the enhancement of the algorithm's accuracy, also increasing the training speed of the model. Pretrained models like Visual Geometry Group (VGG-16) [64], Residual Network (ResNet) [86], and EfficientNet [87] can also be fine-tuned for fault detection using datasets of faulty and healthy panels. An example includes the approach by Kellil et al. [53], who developed a semiautomatic thermal imaging method for localizing and detecting faults in a PV system. The authors compared the accuracy of a small Deep Conventional Neural Network (DCNN) with a fine-tuned model based on VGG-16. The fine-tuned VGG-16 model achieved a slightly higher accuracy when compared to the small DCNN model, reaching accuracies >99% when classifying five different types of faults (i.e., bypass diode failure, partially covered PV module, shading effect, short-circuit, and dust). Moreover, Akram et al. [77] presented an automatic failure detection method using an isolated model and pretrained model, where the data from IR imaging in combination with a CNN managed to achieve a fault detection accuracy of 99.2%. Another approach involved the use of fuzzy-rule base network to classify the degree of severity of each cell of nominal capacity of 40 W using IR thermography [88]. The captured images were compared with similarity with healthy cells and the temperature of the panel was also used for assessing its health status. The degree of severity was split into no, minor, small, medium, major, and critical fault labels. Although the results were promising, there was no evaluation metric that assesses the accuracy of the algorithm.

Preprocessing techniques can also be used to enhance model's robustness and ability to classify noisy images. An example is data augmentation, which uses various image processing techniques, such as cropping and resizing, to improve the model's learning capabilities [89]. The combination of data augmentation with AI enhances the fault detection algorithm's robustness, enabling effective handling of low-quality images [89]. In [90], data augmentation techniques were combined with a CNN model for automatic fault classification. The proposed model yielded a detection rate of 92.5% and a classification rate of 78.85% for eight different types of faults (including soiling, shading, diode, and cracks). Additionally, combining IR with other methods (e.g., EL images) can also enhance the detection of PV faults [63].

To enable faster, cost-effective, and automated monitoring of large-scale PV plants, aerial images, captured by unmanned aerial vehicles (UAVs), could be analyzed [91]. Aerial images provide a comprehensive view of the system and panel's surface, enabling the geolocation and classification of various fault types such as dusty panels, snow-covered panels, panels with physical or electrical damage, and bird droppings. Integrating an IR or a short-wave infrared (SWIR) camera in a drone-based system along with the use of various SVM algorithms was proved to be an effective way for PV monitoring and supervision [92]. In another case, the combination of both aerial visual and EL images was utilized for early fault detection in solar panels, yielding accuracies >91.62% [38]. AI-based models and EL images were also used in [93] to classify faults in solar cells. Another study used EL images to compare the performance of two AI-based techniques, namely SVM and a deep CNN. Trained on 1968 EL images of cells, the CNN model achieved higher fault detection accuracy (88.42%) compared to the SVM (82.44%) [94]. Deep learning models were also utilized for fault detection. In particular, deep learning algorithms were applied to EL images for predicting the PV module power [95]. This approach achieved good accuracy, with over 70% of predictions showing less than 2% error, highlighting its potential for image-based power estimation and fault detection. Finally, Akram et al. [96] developed a lightweight CNN model for automatic defect detection in PV cells using EL imagery. The proposed model achieved over 93% classification accuracy and demonstrated fast inference times, making it suitable for real-time defect identification even on devices with limited computational resources.

2.3.3 | Signal Processing

Signal processing algorithms demonstrate strong capabilities for detecting PV system faults, particularly when integrated with ML methods. This integration facilitates effective anomaly detection and accurate fault classification, significantly enhancing diagnostic precision. Compared to image-based techniques, both methods support fault localization; however, signal analysis typically requires less computational power.

In this area, time domain reflectometry (TDR) analyzes the reflection of electrical signals to identify degradation mechanisms and accurately locate ground faults, open- and short-circuit faults [97, 98]. TDR has been successfully applied to both small- and large-scale PV systems, demonstrating its effectiveness for diagnosing faulty PV strings [99]. In addition to TDR, several other electrical methods are also used for PV fault detection, such as the earth capacitance measurement (ECM) [100] and the wavelet packet transforms (WPT) [73]. ECM measures the capacitance between PV strings and the ground to identify string disconnections and ground faults [101, 102], while WPT analyzes electrical signals in both time and frequency domains to detect and localize bypass diode, line-to-line, and ground faults [36].

Wavelet transforms (WTs) and radial basis function networks were also utilized to detect various faults and classify them as early failures, random, and wear out [103]. The proposed algorithm achieved an efficiency of 98%, taking only 0.2s to execute. Additionally, in another study [104], the use of endowed multiverse optimization, augmented radial basis functional network, and WTs enabled accurate fault detection with errors of 2.5% and execution time of 10 ms. In [105], the isolation forest, continuous wavelet transforms (CWT), and CNN were combined to create an efficient method for identifying anomalies with an overall accuracy of 91%. Furthermore, the discrete WTs using the Haar wavelet, along with the impedance ratio, were successfully utilized to detect, classify, and localize faults in PV systems [36]. Fault occurrences were detected when significant changes in the WT coefficient value of the impedance ratio were observed. To classify the type of faults and precisely locate the faulty string, the voltage of each panel in the string was measured, and a WT was applied to these voltage values. Although Fourier analysis is not commonly used in failure detection approaches, the study by Sevilla-Camacho et al. [106] utilized the discrete Fourier transform on the variations in the alternating current frequency component of PV voltage to detect open-circuit faults, providing accurate results (i.e., 100% effectiveness) in both simulation and field experiments. The developed technique was designed using a simple, low cost and computationally energy efficient setup. Overall, signal analysis methods examine electrical signals for abnormal patterns, offering efficient, real-time fault diagnostics with low computational requirements.

2.3.4 | Summary of Fault Diagnostic Methods

Fault diagnostic methods for PV systems encompass diverse techniques to detect, classify, and localize faults, ensuring system reliability and efficiency. These methods are broadly categorized into electrical data analysis and signal and imaging processing approaches, each with distinct application, advantages, and disadvantages (see Table 2). Those approaches are lately integrated with ML principles to enhance their capabilities for fault diagnostics.

A summary of the articles included in this review that focus on intelligent corrective maintenance methods utilizing the above stated fault diagnostic methods is provided in Table 3, along with their evaluation results.

Based on the reported results, Figure 6 explicitly presents the effectiveness of PV fault detection and classification methods using the accuracy performance metric. Electrical-based solutions achieved median and mean accuracies of 97.1% and 91.0%, respectively, with an interquartile range from 79.8 to 98.6% and a minimum value of 77.8%. Image-based methods showed comparable performance, with a median and mean accuracies of 93.9% and 94.5%, respectively. The minimum value in this case was 88.4%. Signal analysis methods (only three studies reporting the accuracy metric) demonstrated a high mean accuracy of 99.7%, with an interquartile range from 99.5 to 100%. These results indicate that all three diagnostic methods can achieve high accuracy in PV fault detection and classification. Notably, AI techniques have significantly improved the performance of data- and image-based approaches, which rely heavily on pattern recognition capabilities. Among the image-based studies reviewed, only one used a purely statistical method, underscoring the critical role of AI in enabling accurate and robust image analysis. Furthermore, while AI techniques are widely applied to electrical-based methods, statistical methods have also proven highly effective in identifying malfunctions and generating fault detection alerts.

2.3.5 | Reliability Methods

Reliability-based methods evaluate the likelihood and impact of a malfunctioned component or subcomponent on the entire

TABLE 2 | Fault diagnostic methods, techniques, capabilities corresponding detected failures, advantages, and disadvantages.

Method category	Techniques	Capabilities	Detected failures	Advantages (A) and Disadvantages (D)
Electrical data	*I–V*, *P–V* curve analysis, circuit and simulation models, anomaly detection, current/voltage indicators	Fault detection and classification	Mismatch and string faults, inverter faults, open- and short-circuit faults, degradation mechanisms, bypass diode faults, soiling, shading	A: Real-time and remote monitoring, usage of existing monitoring/sensor system D: Limited fault localization, high number of false positives
Image processing	IR thermography, EL, PL, UV fluorescence images, aerial vision	Fault detection, localization, and classification	Module defects and cracks, open- and short-circuit faults, bypass diode faults, degradation mechanisms (e.g., PID), hot-spots, soiling, shading	A: Fast and large-area coverage using aerial means, localization of faulty/defective PV module, high fault detection and classification accuracies D: Dedicated equipment
Signal processing	TDR, ECM, waveform analysis (WPT, CWT), Fourier analysis	Fault detection, localization, and classification	String disconnections, ground faults, line-to-line faults, open- and short-circuit faults, degradation mechanisms, bypass diode faults	A: Low computational power, fault localization D: Intrusive techniques

TABLE 3 | Summary of intelligent corrective maintenance methods.

References	Fault diagnosis method	Input data used	Method type	Faults	Algorithm	Evaluation
[1]	Electrical	Electrical andmeteorological data	AI	Open- and short-circuit, partial shading	Multiple linear regression, random forest classifier	$R^2 = 97.52\%$
[36]	Signal	Electrical signal	Statistical	Line-to-line, line-to-ground, line-to-line-ground, bypass diode failure	The WT technique is used	Accuracy = 100%
[37]	Electrical	Electrical andmeteorological data	AI	Open-circuit, line-to-line	Probabilistic neural network	Accuracy = 100%
[38]	Image	EL images	AI	Soiling, snow coverage, bird droppings, mechanical, electrical damage	DenseNet121 and MobileNetV3	Accuracy = 93.75%
[43]	Image	IR images	Statistical	General	Grayscale conversion, filtering, 3-D temperature representation, probability density function, and cumulative density function	Accuracy = 100%
[47]	Electrical	Electrical and meteorological data	AI	Inverter shutdown, bypass failure and partial shading, open- and short-circuit	DT, k-NN, SVM and FIS	Recall = 91%
[52]	Electrical	Electrical data	AI	Open- and short-circuit, mismatch faults	Logistic regression	Accuracy = 97.11%
[53]	Image	IR images	AI	Bypass diode failure, partial shading, short-circuit, dust deposit	CNN (VGG-16 architecture)	Accuracy = 99.81%
[64]	Image	IR images	AI	Hotspot, hot substring	CNN (VGG-16 architecture)	Accuracy = 98%
[73]	Signal	Electrical signal	Hybrid	General	WPT, ELM	Accuracy = 100% under the no-noise condition, signal-to-noise ratios (SNR) of 30 db = 98.96%, SNR of 35 db = 99.04%, SNR of 40 db = 99.36%.
[69]	Electrical	Electrical data	Statistical	Partial shading	Principal component analysis	Accuracy = 97%
[71]	Electrical	Electrical and meteorological data	Statistical	Open- and short-circuit	Threshold levels	Accuracy = 98.6% and 99.8% for short- and open-circuited PV module failures

(Continues)

TABLE 3 | (Continued)

References	Fault diagnosis method	Input data used	Method type	Faults	Algorithm	Evaluation
[72]	Electrical	Electrical and meteorological data	AI	Inverter faults, network anomalies, MPPT and sensor errors	ELM	Accuracy = 77.83%
[74]	Electrical	Meteorological data	AI	Soiling	ANN	$R^2 = 92.80\%$
[75]	Electrical	Electrical and meteorological data	AI	Soiling	SVM	Accuracy = 80%
[76]	Electrical	Electrical and meteorological data	AI	Bypass diode failure, short-circuit, partial shading	Fully connected neural network	Accuracy = 79%
[77]	Image	IR images	AI	Broken cell interconnection, cracks, failed soldering bonds, glass breakage, localized shunting in cell, high current density at busbars	CNN (both with and without pretrained)	Accuracy = 99.23%
[84]	Image	IR images	AI	Faulty multistring, heated junction box, faulty substring, patchwork pattern, hotspot	CNN	Accuracy = 100%
[86]	Image	IR images	AI	Hotspot, crack, bypass diode activation, offline module, partial shading, soiling, vegetation	CNN (ResNet)	Accuracy = 94.4% (detection), 85.9% (classification)
[87]	Image	IR images	AI	Hotspot, crack, bypass diode activation, offline module, partial shading, soiling, vegetation	Efficientb0 model as its primary foundation	Accuracy = 93.93%
[90]	Image	IR images	AI	Hotspot, crack, bypass diode activation, offline module, partial shading, soiling, vegetation	CNN	Accuracy = 92.5% (detection), 78.85% (classification)
[93]	Image and electrical	EL images and I–V curves	AI	Soldering faults, manufacturing defects, broken cell, partial shading, crack	CNN, SVM, k-means, random forest, multilayer perceptron	Accuracy = 81.60%

(Continues)

TABLE 3 | (Continued)

References	Fault diagnosis method	Input data used	Method type	Faults	Algorithm	Evaluation
[94]	Image	EL images	AI	Cracks, cells with electrically separated and degraded parts, short-circuit, interconnection failure, soldering failures	CNN, SVM	Accuracy = 88.42% (CNN) 82.44% (SVM)
[96]	Image	EL images	AI	Crack, gridline, inactive	CNN	Accuracy = 93.02%
[103]	Electrical	Electrical and meteorological data	AI	Short-circuit, partial shading, abnormal aging, hybrid faults	CNN and residual-gated recurrent unit (Res-GRU)	Accuracy = 98.61%
[104]	Electrical	Electrical and meteorological data	Hybrid	Open- and short-circuit, partial shading	Endowed multiverse optimization (EMVO) methodology for PV power prediction, augmented radial basis functional network (ARBFN) technique is implemented for exact fault detection	Accuracy = 98%
[105]	Signal	Electrical signal	Hybrid	Line-to-ground, line-to-line, three-phase	Isolation forest technique to detect the fault, CWT for wave analysis, CNN for classification	Recall = 96%
[106]	Signal	Electrical signal	Statistical	Open-circuit	Discrete Fourier transform and a classifier based on statistical techniques.	Accuracy = 100%

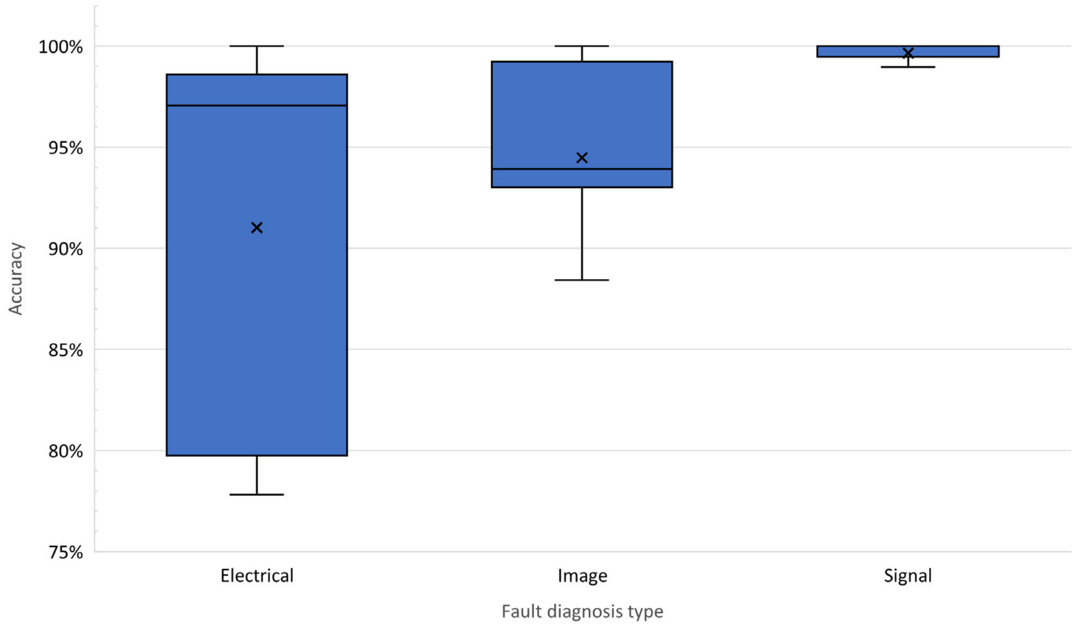

FIGURE 6 | Box plot showing the accuracies reported by the different fault diagnostic methods.

system [107], making them essential for system planning and long-term operation. By enabling effective risk assessment and identification of critical components, these methods help minimize revenue losses through informed decision-making. Additionally, they assess the economic implications of failures to tailor maintenance strategies, facilitating the prioritization and scheduling of field actions. As demonstrated by Livera et al. [7, 67] and Lindig et al. [108], DSS based on reliability-centered approaches offer data analysis, reporting, and visualization capabilities. They also provide quantitative assessments of issues and recommend field actions to enhance performance and minimize energy losses. Additionally, the remaining lifetime of each equipment could be determined by considering historical data such as the failure rate of each component. In this context, the exponential and Weibull distributions are commonly used to simulate the reliability functions of different PV system components, aiding in the estimation of failure probability, and remaining useful lifetime [109, 110].

Reliability methods also enable the identification of the most critical components within a PV system [111]. For example, field-derived datasets have shown that transformer and inverter issues account for almost two thirds of total energy losses in PV systems [111]. To assess the criticality of each component, several methods are used, including failure mode and effects analysis (FMEA) [28], multicriteria decision analysis (MCDA) [112], reliability, availability, and maintainability (RAM) [113] analysis, and the cost priority number (CPN) method [8]. An example of CPN analysis is shown in Figure 7, where the cost of repair and the lost production per year for various cases were investigated. Some cases, such as improper installation, resulted in low annual production losses but high overall repair costs, while others, like soiling, had low repair costs but high production losses. By applying the CPN method, PV operators can decide which maintenance actions to prioritize to minimize lost production due to faulty components.

FMEA is another well-established semiqualitative method that systematically examines each component to identify potential failure modes and assess their impact on the overall system functionality. In the context of PV systems, FMEA is used to pinpoint critical components and failure modes that could compromise

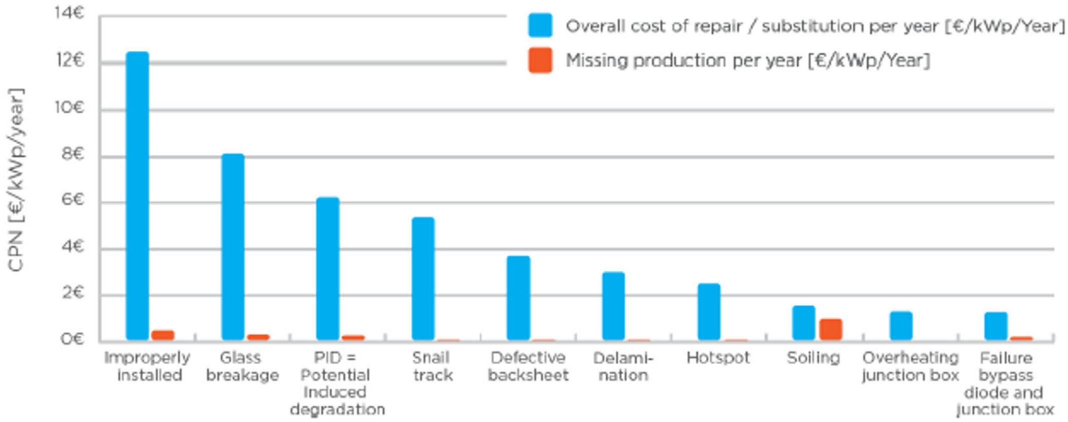

FIGURE 7 | CPN analysis and energy loss for the 10 most common faults for PV modules. Figure was obtained from [28].

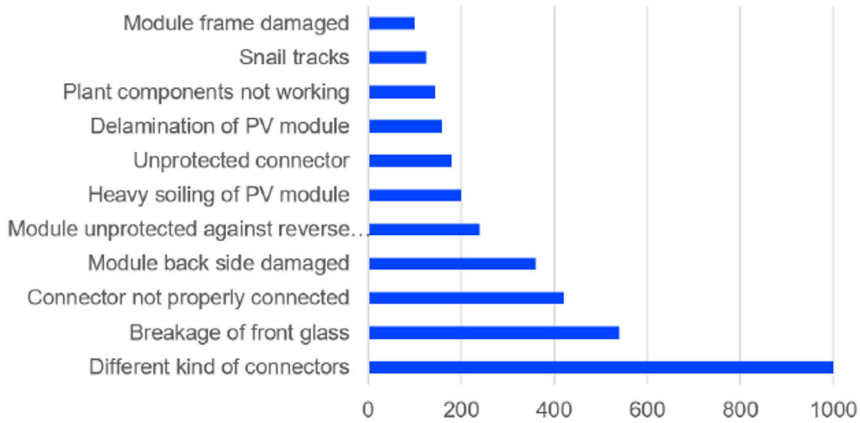

FIGURE 8 | FMEA rating of PV module failures. Figure was obtained from [28].

system performance and reliability. Each identified risk is evaluated for its severity (S), occurrence (O), and detectability (D). Figure 8 illustrates an example of FMEA rating of PV module failures, highlighting that various types of connectors, front glass breakage, and improper connector connections pose the highest risks to system performance.

MCDA assesses the effect of different actions in the system, helping PV operators/owners to optimize their revenue and it can be used for selecting the parameters that are most important for selecting a PV site [112]. The failure mode effects criticality analysis (FMECA) and the Markov process can also be utilized to predict the probability of failure of different balance of system components [114, 115]. It was found that the inverter and ground system were the most critical components for maintaining the normal operation of the PV system. Between 2010 and 2012, a survey of 350 PV installations [110] showed that inverters were the dominant source of energy loss, responsible for about 36% of the total yield reduction. Also, AC subsystem issues and externally caused outages comprise a large share of the biggest losses (40% in total). On the other hand, PV module failures represented a small fraction (1%) of identified issues. Bansal et al. [116] also used the FMECA method to analyze degradation and failures in a PV power plant. The study evaluated O&M approaches, reliability evaluation and module degradation, pointing out that ethylene vinyl acetate (EVA) discoloration, snail trails, back sheet burning, and glass breakage were identified as the most common and severe defects and failures in PV modules. Regarding the failure causes, humidity and repetitive thermal cycling impose the greatest stresses on power-electronic components [117], whereas PV modules suffer from increased degradation in regions with high temperatures and UV intensities [118]. Cables can also be the cause of excessive stress experienced by the system during its operation. Moreover, Livera et al. [67, 119] utilized the FMECA approach to assign criticality value to the detected failure incidents and generate recommendations for resolving the underperformance incidents in PV systems. Finally, Moser et al. [8] used a combination of a cost based FMEA and CPN to assess the criticality of each underperformance issue based on their economic impact, including major issues in PV modules and inverters.

A summary of the reliability methods reviewed in this study is provided in Table 4.

Overall, reliability methods (integrated within DSS) can provide useful insights to technicians concerning the most frequent and critical failure mechanisms that occur in PV modules and power electronic devices to optimally schedule the field maintenance actions. However, the selection of the method depends on the driving factors, e.g., data availability and the level of technology maturity as shown in Figure 9. Literature lacks a comparative study of the most robust analytical reliability method for assessing incidents' criticality and optimizing the field O&M interventions.

2.4 | Predictive Maintenance

Predictive maintenance is a condition-based intervention carried out following a forecast-based analysis and a continuous evaluation of the system's conditions (e.g., health state of the PV system) [120]. As such, appropriate equipment (e.g., database to access historical data, sensors to collect real-time operational data from a system), monitoring software, and advanced analytics (to analyze the collected data, identify trends/patterns, and predict potential failures) are required to enable timely interventions. Predictive maintenance utilizes anomaly detection, fault diagnosis, and forecasting to mathematically evaluate the risk of failure of system components and anticipate equipment failures and deterioration in advance. This allows to better plan any required field maintenance activities, while also enabling the detection of potential failures at early stages or before they occur/escalate [49, 121–123]. Predictive maintenance thus reduces unplanned downtime and energy losses, lowers the cost of post maintenance, and prevents "bigger" failures (e.g., technicians can fix the problem before the damage gets bigger). The combination with prescriptive analytics enables optimal decision-making regarding maintenance activities. These analytics support both proactive and corrective measures, allowing interventions to be scheduled based on real-time conditions and forecasts rather than fixed intervals.

Even though the implementation of predictive maintenance strategies could have a positive impact on reducing the LCOE, the implementation of these strategies is still not so common in the PV sector. The lack of maintenance logs needed for the validation of predictive maintenance algorithms makes the implementation of data-driven techniques more difficult.

TABLE 4 | Summary of reliability methods reviewed in this article.

References	Method	Use Case	Results
[8]	FMEA and CPN	Economic impact of technical risks and failures	For PV modules, common failures like glass breakage, EVA discoloration or defective back sheets bear a higher level of economic risk. Unlike the PV module risks, the inverter-related risks appear to have a significant impact on production.Wrong/absent cables connection and broken/burned connectors are two of the most detrimental (economically technical risks found).
[28]	CPN	Economic impact of technical risks	Improper installation, glass breakage and PID were found to be the technical risks with the most economic impact.
	FMEA	Rating technical risks based on failure reports	Different kinds of connectors, breakage of front glass and not proper connection of the connector had the highest rating in causing PV module failures.
[67]	FMEA	DSS	Reduced response and resolution times could improve the PV power production of the test PV plant by up to 2.41%. Even for 1% energy yield improvement by performing corrective actions, a DSS is recommended for PV plants with at least 250 kWp capacity.
[111]	FMEA	Rank failures in different subsystems	Inverter and transformer were found to cause most of the issues.
[112]	MCDA	Determine ideal PV site location	Solar radiation, temperature, and distance to transmission line were found to be the most important parameters for selecting site location.
[113]	RAM	Identification of weak links to improve system reliability	Calculated the expected lifetime and the best probability density function for each component.
[114]	FMECA	Reliability assessment of PV system	Inverter and ground system were the most critical components for maintaining the normal operation of the PV system.
[115]	FMEA	Reliability assessment of PV system	PV inverter has a high RPN value and it was found to be the most critical component.
[116]	FMECA	Reliability assessment of modules after 10 years	EVA discoloration, snail trails, back sheet burning, and glass breakage are identified as the most common and severe defects and failures observed in PV modules.
[119]	FMECA	DSS	Approximately 7% of lost energy production could be recovered by performing field mitigation activities.

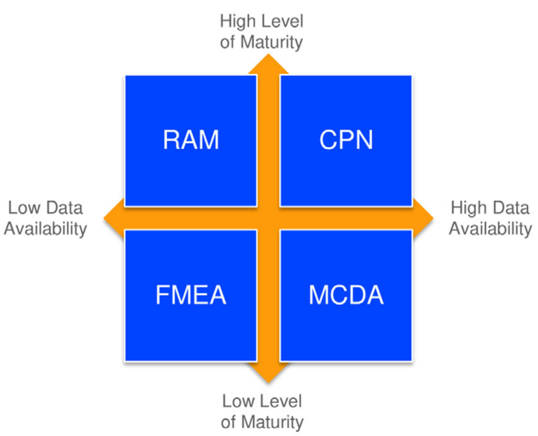

FIGURE 9 | Selection guide for different methods based on the driving-factors: data availability and level of technology maturity. Figure obtained from [28].

Additionally, AI algorithms used in predictive maintenance include the application of supervised learning algorithms, which require labeled training datasets to predict failure modes. Such labeled datasets are usually not available for existing PV installations. Moreover, the sudden and instantaneous nature of various malfunctions as well as the nonlinearity and lack of periodicity in the measured data makes the implementation of predictive maintenance strategies and prediction of future faults even more challenging [124]. Finally, the effectiveness of AI-driven predictive maintenance models relies heavily on the availability, quality, and diversity of data sources. Despite the difficulties in developing predictive maintenance models, there is a growing need to tackle the challenging problem of intelligent predictive maintenance strategies in PV systems. This surge in research is likely driven by the rapid expansion of PV system installations and the increasing need for autonomous algorithms that can address the operational challenges of PV systems by generating predictive maintenance alerts.

In this field, Laguna et al. [125] proposed a recursive least squares algorithm to predict PV plants' energy generation and indicate abnormal behavior. The study forecasted long-term anomalies using historical field data. The efficacy of the AI algorithm was validated in 20 PV systems (ranging from 30 kWp to 300 kWp) in Spain, facilitating long-term diagnosis and degradation assessment. Liu et al. [126] developed a data-driven method for lifetime prediction of PV arrays. This method was based on a nonlinear Gamma stochastic process to estimate the probability function of the degradation trend. The Gamma degradation process is an ideal candidate for modeling slow degradation processes, as indicated in [127]. The researchers utilized the performance ratio (PR) and the degradation trend to extract the remaining useful life of power devices. An actual time of failure for the PV array equal to 6.38 years was found, which was about 17 days more than the actual failure time. Theristis et al. [128] proposed a degradation model that incorporated a nonlinear technique, to extract the long-term PV degradation rate and determine its financial impact on LCOE.

Apart from predicting the long-term performance losses in PV systems, research and industrial efforts also focused on the development of algorithms for predicting PV underperformance incidents. An example of a predictive maintenance algorithm includes the study conducted by Betti et al. [50], who utilized a self-organizing map (SOM) to predict the emergence of faults over a forthcoming 7-day period, providing generic fault alerts in advance. The proposed data-driven approach generated alerts on average up to 7 days in advance, achieving true positives rate up to 95% for incipient faults. In [129], the K-means clustering method and the Long Short-Term Memory (LSTM) algorithm were utilized to predict anomalies in current measurements and to generate alerts for predictive maintenance actions. Furthermore, Huuhtanen and Jung [130] implemented a CNN network for generating predictive maintenance alerts based on the comparison of the daily electrical power curve with neighboring panels. The deep learning method managed to outperform the average of the two adjacent panels prediction and detected a shading incidence that was occurring every day between 10:00 am to 12:00 pm. The algorithm was validated on both synthetic and real data showing exceptional accuracy with mean squared error (MSE) error <1%. However, the dataset of the given method was very limited, and therefore more data are needed for validating the ability of the algorithm to predict faults. In another study [123], several ensemble ML algorithms (i.e., random forest, XGBoost, CatBoost, and LightGBM) were used to determine predictive maintenance needs in a 99.9 kWp PV system. The XGBoost algorithm exhibited the highest accuracy (98.62%) for predicting the maintenance needs [123].

A recent study has also shown the potential of identifying inverter behavior malfunctions several weeks prior to a failure occurring [122]. Specifically, the study used ML algorithms along with a multivariate residual analysis for comparing the expected against the actual values from data channels. Expected values were derived using the SOM algorithm and auto-encoders. Control charts and Hotelling's test were then applied to trigger alarms when significant deviations were detected. The model proved capable of identifying up to 83% of failures a week before an inverter fails [122]. Moreover, Khalil et al. [131] used a robust deep-learning transformer model to forecast faults (such as short-circuit, open-circuit, and shadow) in PV systems. The algorithm achieved a mean average error (MAE) of 0.09377 when forecasting faults. A drawback of this study was that the results were verified only in a simulated model developed in MATLAB. Vyas et al. [132] managed to forecast the maintenance issues (e.g., grid and inverter failures) for a PV power plant using random forest and vector autoregression methods. The developed model had a forecasting error (root mean squared percentage error—RMSPE) <10%.

An approach for predicting the health status of PV systems was proposed in [133] using correlation, clustering, and feature extraction methods. The approach was validated with real current data from a PV array composed of twelve modules (power between 205 and 240 Wp) in three health states, namely broken glass, healthy, and big snail trails. The obtained results showed the efficacy of the approach for predicting the three health states and determining the PV module degradation level that indicates priorities to corrective and predictive maintenance actions. The authors also claim that this is a generic and cost-effective solution, since it uses only electrical measurements that are already available in standard data acquisition systems. In [134], a warning level system based on SVM One-Class Rule algorithm and the confidence interval of the forecasted temperature-corrected PR trend of a PV system was developed to generate both fault detection and predictive maintenance alerts. The algorithm achieved a sensitivity of 90.9% for predictive maintenance alerts. However, the proposed algorithm was validated on a single PV plant.

In another study [135], a data-driven predictive maintenance alerting routine for utility-scale PV systems was introduced. The proposed methodology leverages a ML model for PV performance prediction along with statistical thresholds for fault detection and predictive maintenance. The XGBoost algorithm was used to predict AC power and PV system underperformance. The proposed system achieved a sensitivity of 93.9% for fault detection and 83.3% for fault prediction up to 10 days in advance, demonstrating its effectiveness in supporting proactive maintenance scheduling. Similarly, Livera et al. [136] presented a predictive maintenance framework for PV systems based on the XGBoost algorithm. The model was developed to predict AC/DC power at the inverter level and to generate daily predictive alerts based on residuals analysis, achieving a fault prediction of 87.5% when validated against real maintenance logs.

Montes Romero et al. [137] have developed a trend-based predictive maintenance routine for utility-scale PV systems, combining time series analytics, PR monitoring, and statistical hypothesis testing to detect PV underperformance issues (such as PID, shading, clipping, and degradation). The methodology evaluates deviations in PR, voltage slope, and insulation resistance to identify/predict PID conditions and to forecast insulation resistance degradation. The routine was capable to predict PID loss decreases and forecast low insulation incidences, though it was validated using simulated/imputed data. Moreover, a fault prediction and health status assessment framework for PV inverters, combining WT with a deep neural network (DNN), was proposed in [138]. A Health Status Index (HSI) was constructed from the DNN output, and its trend was predicted using an ARIMA model to enable early fault warnings. The system was trained using historical fault and normal operation data and achieved an accuracy of 95.8%, with a 2.1% false alarm rate, and 1.9% missed report rate.

In parallel, private companies (e.g., SmartHelio [11]) also developed predictive maintenance algorithms. For example, SmartHelio developed an algorithm that enables real-time fault prediction and localization. The developed model uses IoT-based hardware combined with ML techniques to improve data-driven fault detection and prediction in PV systems [139]. The SmartHelio solution could substantially decrease underperformance losses and O&M costs through predictive analytics [11]. Moreover, predictive maintenance in PV systems has been successfully demonstrated through a hybrid AI model developed as part of the Innosuisse project [140]. This model combines data-driven AI techniques with domain-specific physical knowledge, referred to as physics-informed AI, to support optimized maintenance strategies. Implemented in collaboration with Fluence Energy, the system detects energy loss events stemming from factors such as soiling, electronic component failures, and tracking system malfunctions, while also determining the most suitable timing for maintenance actions. By integrating historical operational data with physics-based insights, the model generates realistic fault scenarios to train diagnostic algorithms. This enables early fault detection and facilitates cost-effective interventions, ultimately reducing energy losses, improving system reliability, and promoting more efficient solar plant operations.

A summary of all predictive maintenance approaches, along with their capabilities and evaluation, reported in the literature is provided in Table 5. Studies lacking evaluation metrics, details about the developed predictive algorithm, and information about the input features were excluded.

From the table, it is evident that most of the developed PV predictive maintenance algorithms focused on large PV systems (>100 kWp) and combined statistical methods with AI techniques (that are increasingly adopted lately) for fault prediction. The reported studies used the accuracy metric more frequently and achieved accuracies >80% (ranging from 83.0 to 99.6%, with a median of 97.2%). The sensitivity and recall metrics were used occasionally, with the literature studies reporting sensitivities >83% and recall >87%, underscoring the algorithms capabilities for generating predictive maintenance alerts. Hybrid (AI and statistical) methods yielded accuracies up to 99.6%, while AI and statistical techniques achieved accuracies up to 98.6% and 83%, respectively, though direct comparison of the different statistical/AI methods is difficult to be done since the reported studies used different performance evaluation metrics (i.e., accuracy, sensitivity, and recall) and test setups. In summary, almost all the studies presented in the field focused on developing algorithms to generate predictive maintenance alerts. Most of the developed algorithms are site specific (tested and validated only on one PV system), with no proven applicability to PV plants located in diverse locations and climatic conditions. This limitation hinders the generalization and the universal applicability of such algorithms across different PV installations, setups, and PV module technologies.

2.5 | Extraordinary Maintenance

Extraordinary (or urgent case) maintenance occurs when major unpredictable events occur at a PV power plant, necessitating immediate actions to restore normal PV operation. Such interventions are typically triggered by force majeure event (e.g., flooding, hurricanes, tornados, hail, and lightning), theft, fire, etc. and when modifications are required by regulatory changes. Extraordinary maintenance refers to nonroutine repairs and interventions prompted by unexpected failures, severe weather damage, or accelerated degradation.

A literature survey revealed a comparative analysis study that demonstrated modest median short-term production loss (~1% of annual output per incident) from extreme weather conditions. However, outliers can result in losses as high as 60%, underscoring the severe risk posed by extreme conditions [143]. Hail events with diameters of 25 mm or greater were associated with elevated degradation rates (performance loss rates, PLRs), even in modules certified under IEC 61 215 for similar impact thresholds, suggesting the need for more stringent testing protocols. Wind speeds exceeding 90 km/h were similarly correlated with increased PLRs, indicating that both wind resilience and installation best practices (e.g., mounting configurations, shielding from nearby structures) are critical to minimizing damage [144, 145].

To address the increasing risks posed by extreme weather events, PV systems must take into account the risk assessments of each site location that account for localized climate hazards such as hail, hurricanes, heatwaves, and blizzards. The industry must develop weather forecasting tools and mitigation approaches that will enable the reduction of losses and damages to PV system during extreme weather conditions. Enhanced data collection through IoT and weather sensors, including centralized reporting on damage types, repair costs, and long-term impacts on performance and LCOE will enable operators and technicians to get valuable insights.

Due to the unpredictability of these events, AI has not yet been widely utilized in literature for extraordinary maintenance. However, advancements in AI-driven predictive analytics, weather forecasting, and automated diagnostics are expected to expand the role of extraordinary maintenance in addressing such challenges. AI-powered weather forecasting systems can increasingly predict the likelihood of extreme events (such as hailstorms, sandstorms, or hurricanes) with growing accuracy. By integrating local meteorological data, terrain characteristics, and historical storm patterns, these models enable site-specific risk assessments for PV installations. This predictive capability supports proactive mitigation measures before adverse weather strikes, reducing the risk of mechanical damage and long-term performance degradation.

After a storm, drones equipped with high-resolution imaging (IR and EL) and AI-based image processing algorithms can be deployed for efficient, accurate post-event inspections. These technologies facilitate the early identification of faults such as cracks, enabling timely maintenance interventions and minimizing energy yield losses.

TABLE 5 | Summary of predictive maintenance approaches reviewed in this article. PV system sizes are labeled as small (<10 kWp), medium (10–100 kWp), and large (>100 kWp).

References	Input features	Time interval	PV system size	Data type	Method	Faults	Evaluation
[11]	Electrical and meteorological data (I_A, V_A, G_I, T_{inv})	–	Large	Real-time	Hybrid	Various underperformance issues on inverter, module and sensor level	Accuracy = 99.6%
[50]	Electrical and meteorological data (T_{inv}, P_{out}, GHI)	5 min	Large	Historical	AI	Overvoltage, ground fault, sensor faults	Recall = 95%
[141]	Meteorological data (T_{inv}, T_{amb}, GHI)	Hourly	Large	Historical	AI	–	$R^2 = 0.89$ (direct), $R^2 = 0.62$ (survival)
[123]	Electrical data (V_A, I_A)	–	Large	Historical	AI	Maintenance needs	Accuracy = 98.62%
[130]	Electrical data (P_A)	Hourly	Small	Historical	AI	Underperformance	MSE = 0.00001 (synthetic), 0.00235 (real)
[131]	Electrical and meteorological data (I_{SC}, V_{OC}, V_A, I_A)	–	Small	Synthetic	AI	Shading, overlapping, open- and short-circuit	MAE = 0.0938
[132]	Electrical data (P_{out}) and maintenance log	Daily	Large	Historical	Hybrid	Grid failure, inverter failure, module cleaning and cloudy	RSMPE < 10%
[133]	Electrical data (I_A, I_{SC})	1 min	Small	Historical	Hybrid	Health status	F-value = 87.5%
[134]	Electrical and meteorological data (I_A, V_A, PR, G_I, T_{inv})	15 min	Large	Historical	Hybrid	Underperformance	Sensitivity = 96.9% (fault detection), Sensitivity = 92.9% (predictive maintenance)
[135]	Electrical data (P_{out})	Hourly	Large	Historical	Hybrid	Underperformance	Sensitivity = 93.9% (fault detection), Sensitivity = 83.3% (predictive maintenance)
[136]	Electrical and meteorological data (G_I, T_{mod}, P_{out}, P_A, T_{amb})	Hourly, daily	Large	Historical	Hybrid	Underperformance	Recall = 87.5%
[138]	Electrical and meteorological data (V_A, I_A, V_{out}, I_{out}, P_A, T_{mod})	–	–	Historical	Hybrid	Inverter failures	Accuracy = 95.8%
[142]	Electrical and meteorological data (I_A, V_A, G_I, T_{inv}, T_{mod})	–	Large	Historical	Statistical	Inverter failures	Accuracy = 83%

[a]AC output current (I_{out}), AC output voltage (V_{out}), AC output power (P_{out}), ambient temperature (T_{amb}), array DC current (I_A), array DC voltage (V_A), array DC power (P_A), Global Horizontal Irradiance (GHI), in-plane irradiance (G_I), inverter temperature (T_{inv}), module temperature (T_{mod}), open-circuit voltage (V_{OC}), performance ratio (PR), short-circuit current (I_{SC}).

3 | Discussion and Recommendations to Improve Solar Asset Management

3.1 | Requirements for Intelligent PV Predictive Maintenance

Effective predictive maintenance in PV systems relies on several key requirements (e.g., robust data collection infrastructure and advanced sensor network and analytics). Foremost among these is a reliable Supervisory Control and Data Acquisition (SCADA) system, supported by an advanced sensor network, consisting of both electrical and meteorological sensors. This infrastructure enables the continuous, high-frequency collection of real-time data on PV performance, electrical parameters, and environmental conditions [39]. Access to detailed maintenance logs and high-resolution historical (ideally labeled) data is also a critical requirement for developing accurate data-driven predictive maintenance algorithms. Moreover, the integration of reliable communication protocols, interoperability interfaces, and cloud infrastructure is necessary to ensure seamless data collection, storage, and analysis. Advanced technologies such as IoT sensors, smart meters, AIoT, edge computing devices, and DSS further enhance these capabilities. Finally, intelligent analytics powered by AI/ML techniques are crucial for processing real-time and historical data from various sources, allowing the early detection and prediction of failure patterns in critical components.

In addition to the above requirements, PV predictive maintenance can also utilize other diagnostic methods, such as image processing. In such cases, appropriate equipment (e.g., thermal and EL cameras, UAVs and/or drones) for obtaining images and deep learning techniques for processing is also required [40].

3.2 | Availability and Quality of Measurements for Intelligent Data-Driven PV Predictive Maintenance

The effectiveness of data- and AI-driven predictive maintenance is heavily dependent on the availability and quality of data [78]. AI models require a continuous flow of real-time and historical data from SCADA. In cases of data unavailability, even synthetic data can be generated and used to train the AI models [146]. A review of the literature on PV predictive maintenance revealed that historical data were the predominant source used in 79% of the reviewed studies, while synthetic and real-time data accounted for 14% and 7%, respectively. This highlights a strong dependence on previously recorded data for model development and fault prediction.

In parallel, the accuracy of AI-based models is improved when high-resolution datasets (covering a long evaluation period) are available, enabling them to detect complex patterns. Insufficient, missing, or incomplete datasets can influence the accuracy of the predictions and decision-making, while also introducing bias to the results [78, 79]. Regarding the quality of data, noisy or inaccurate data can lead to false predictions, either missing occurred and upcoming failures (false negatives) or generating unnecessary maintenance alerts (false positives) [78, 79].

3.3 | Input Feature Selection for Intelligent Data-Driven PV Predictive Maintenance

Input feature selection for intelligent data-driven predictive maintenance in PV systems involves the identification of critical parameters that enable accurate forecasting of fault conditions and performance degradation modeling. Key input features include the main PV electrical (e.g., current, voltage, and power) and environmental measurements (such as irradiance, ambient temperature, and module temperature). Apart from the main monitored electrical and meteorological data, other monitored weather data (e.g., wind speed and direction, relative humidity, rainfall, and snow) and parameters such as weather forecasts, PV performance trends and parameters (e.g., PR, soiling ratio), KPIs and criticality evaluation parameters, inverter and PV module manufacturer specifications, and plant's capacity and location (geographical coordinates) along with historical data and maintenance log files could also be used as input features in the predictive algorithms' development phase.

From the literature survey of the developed predictive maintenance algorithms, DC current, DC voltage, and AC output power measurements were the most frequently used electrical data, followed by key meteorological variables such as in-plane irradiance, inverter and module temperature (see Figure 10a). When broadly grouped into measured parameters (see Figure 10b), temperature, current, and power measurements stand out as the most frequently used features for PV predictive maintenance. These findings highlight the critical importance of

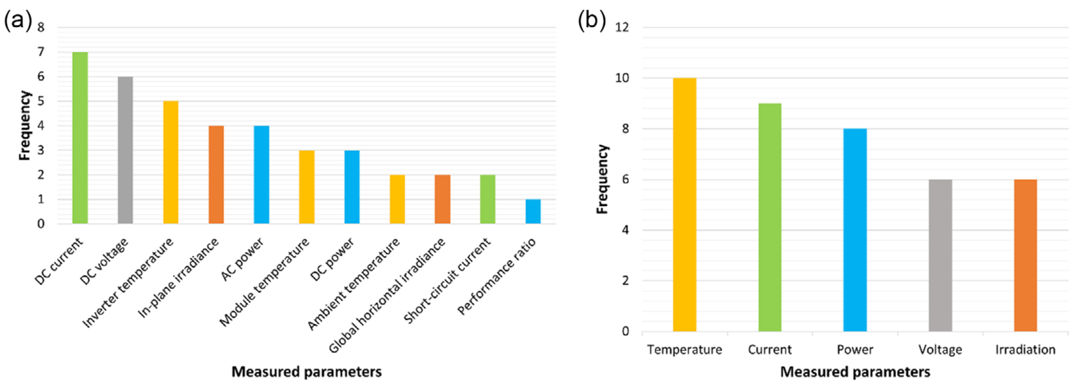

FIGURE 10 | (a) Frequency of input features used for PV predictive maintenance. In (b), the features are broadly grouped into measured parameters.

integrating both electrical and environmental parameters for the development of predictive maintenance algorithms. Also, such features can be used to improve the accuracy of the decision system for generating maintenance alerts.

3.4 | Modern Technologies Integration, Key Benefits, and Predictive Maintenance Capabilities

The integration of modern technologies into PV systems is transforming the O&M field, enabling remote and real-time monitoring, early fault diagnostics, and accurate fault prediction. Key technologies such as IoT, AI, ML, digital twins, edge computing, and big data analytics play a crucial role in making PV predictive maintenance more efficient and cost-effective. These innovations allow operators to anticipate failures before they occur, reducing downtime periods and operational costs, improving performance, and increasing PV energy output.

More specifically, IoT and advanced sensors form the foundation of predictive maintenance by continuously monitoring PV electrical and meteorological measurements along with key performance metrics. These sensors collect real-time data with enhanced precision [147] and transmit it to a central monitoring system via cloud or edge computing. AI-powered models can be then integrated to analyze historical/real-time measurements, maintenance log files, performance data, and emerging trends to detect anomalies/patterns reliably and predict accurately future failures/potential PV underperformance issues. The use of AI-driven analytics improves the accuracy of fault detection/prediction, thus minimizing false alarms, while also enabling the provision of field maintenance actions to enhance the PV performance.

Digital twin technology further enhances predictive maintenance by creating a virtual replica of the PV system, allowing operators to simulate different conditions and test maintenance strategies without disrupting operations. The synchronization of the digital twin technology with live data streams provides real-time analytics and predictive insights that can be used by PV plant operators to enhance decision-making.

Edge computing and cloud computing provide the necessary infrastructure for storing and processing vast amounts of data from sensors and historical records to predict equipment failures before they occur. Edge computing enables faster decision-making by analyzing critical data locally, reducing the latency associated with cloud-based processing. Meanwhile, cloud computing supports large-scale data analytics, enabling predictive models to improve over time. The edge devices analyze sensor and inverter data on-site and transmit only the relevant insights to the cloud. Consequently, the use of edge computing and cloud computing in predictive maintenance enables faster response times, reduced network congestion, and improved real-time monitoring and supervision.

Big data analytics (that use AI and ML algorithms to process structured and unstructured data from PV systems) help to identify long-term PV degradation and trend-based performance losses. Finally, blockchain technology contributes to smart PV maintenance by ensuring data security, transparency, and tamper-proof record-keeping. Smart contracts can automate maintenance actions based on AI-driven insights and real-time analytics, streamlining operational efficiency.

The benefits of integrating modern and advanced technologies into PV predictive maintenance are substantial. Predictive maintenance reduces operational costs by preventing costly unexpected failures and scheduling maintenance proactively, reducing unplanned downtime. It also enhances annual energy yield by ensuring that PV modules and inverters operate at optimal levels. Extending equipment lifespan is another key advantage, as early detection of failures and patterns allows timely interventions. Additionally, remote monitoring and diagnostics improve system reliability, reducing the need for manual inspections [147]. The integration of AI and big data analytics for fault detection and prediction provides real-time analytics and actionable insights, enabling quick troubleshooting and optimal decision-making for improving performance.

Smart PV monitoring systems equipped with predictive maintenance capabilities can detect anomalies, predict failures, and automate maintenance scheduling based on system operational status and health state. These systems classify issues such as shading, soiling, and degradation, allowing operators to take corrective and proactive actions before performance is significantly impacted. Remote monitoring enables real-time tracking through cloud-based dashboards and mobile applications, reducing the need for on-site inspections. AI forecasting tools can be used to forecast the PV generation and adjust the maintenance interventions (e.g., cleaning schedules). Additionally, intelligent monitoring systems enhance grid interaction by monitoring power quality and voltage fluctuations, ensuring stable energy distribution and support demand-response strategies.

As the energy sector embraces digitalization, smart predictive maintenance is becoming increasingly vital for PV systems. The convergence of AI analytics and IoT technologies have the potential to transform the future of maintenance in the solar industry by analyzing vast amounts of data from sensors, detecting failures, identifying patterns, and generating predictive maintenance alerts. This integration of AI and IoT enables a shift from reactive to proactive maintenance, where potential issues are addressed before they escalate into significant problems (e.g., intervention before a breakdown occurs), preventing costly disruptions. The result of this synergistic approach is increased PV performance, improved equipment reliability, reduced unplanned downtime, and lower maintenance and repair costs. The transformative advantages of AI in IoT for the solar industry are summarized in Figure 11.

3.5 | The Cost Benefits of Predictive Maintenance

Even though incorporating predictive maintenance strategies has higher upfront costs compared to corrective maintenance [148], the cost savings and performance improvements over the PV system's lifetime make it a profitable investment. More specifically, integrating predictive maintenance into solar installations offers significant financial benefits by lowering maintenance expenses, while also enhancing PV system's energy yield. By avoiding unexpected equipment failures, which often lead to costly

FIGURE 11 | The advantages of the new paradigm known as the AIoT for the solar industry.

emergency repairs and production losses, through optimal maintenance scheduling, PV plant owners could reduce the O&M costs by up to 35% compared to reactive maintenance strategies as shown by different case studies [11, 142, 149]. This proactive strategy enables technicians and operators to handle potential issues during scheduled downtimes rather than reacting to sudden breakdowns. Predictive maintenance also improves the uptime of the system typically by 9% [142] while also decreasing the spare parts inventory costs. Furthermore, enhanced energy production through well-maintained systems can increase the lifetime of aging assets and annual revenue by up to 20% [142] and 5% [142], respectively, making predictive maintenance a financially sound investment for PV system owners.

3.6 | Standardized O&M Framework and Future Directions to Improve Solar Asset Management

Establishing standardized frameworks for intelligent PV O&M is crucial for enhancing solar asset management. The International Energy Agency Photovoltaic Power Systems Programme (IEA-PVPS) underscored the importance of such standardization in its O&M guidelines [150], highlighting that standardized O&M services are essential for effective PV plant management. The standardized O&M framework should incorporate cutting-edge technologies (e.g., IoT sensors and AI-driven algorithms) along with interoperability protocols and cloud computing, bid data and advanced analytics (descriptive, diagnostic, predictive, and prescriptive), and cost-based decision-making systems to improve solar asset management. By following such a standardized framework, PV plant operators/owners can schedule the maintenance activities and perform the appropriate field actions, enhancing asset long-term performance and sustainability.

Based on the main findings of the present study and the discussion reported in Section 3, the framework for implementing smart maintenance strategies should involve the following structured series of steps (see Figure 12): (a) data acquisition system (DAQ), (b) data preprocessing and descriptive analytics, (c) diagnostic analytics, (d) predictive analytics, and (e) prescriptive analytics.

In parallel, the integration of web-based databases and structured tools such as the TRUST-PV Risk Matrix and its associated Solution Matrix [151] along with DSS [7, 67, 108, 151] to the comprehensive framework for smart maintenance strategies can help standardize the assessment of O&M events and can provide the appropriate recommendations for corrective, predictive, or preventive actions to be scheduled based on forecasts and economic and performance impact. Their altogether combination will minimize system downtime and increase the overall PV performance.

To align with the current trends in machine learning, the integration of generative AI-driven tools is strongly recommended for smart O&M in PV systems [152]. Generative AI, particularly models based on large language models (LLMs) such as generative pretrained transformer (GPT), excels at organizing and synthesizing data from diverse sources, recommending optimal analysis tools, and establishing meaningful connections between disparate data streams. In practical terms, these tools can deliver tailored insights, automated analysis and reporting, enhanced predictive maintenance, insight-driven asset management, and support strategic decision-making for O&M actions, ultimately improving revenues for PV plant owners [25]. However, the successful implementation of generative AI requires attention to workforce upskilling, ongoing digital infrastructure upgrades,

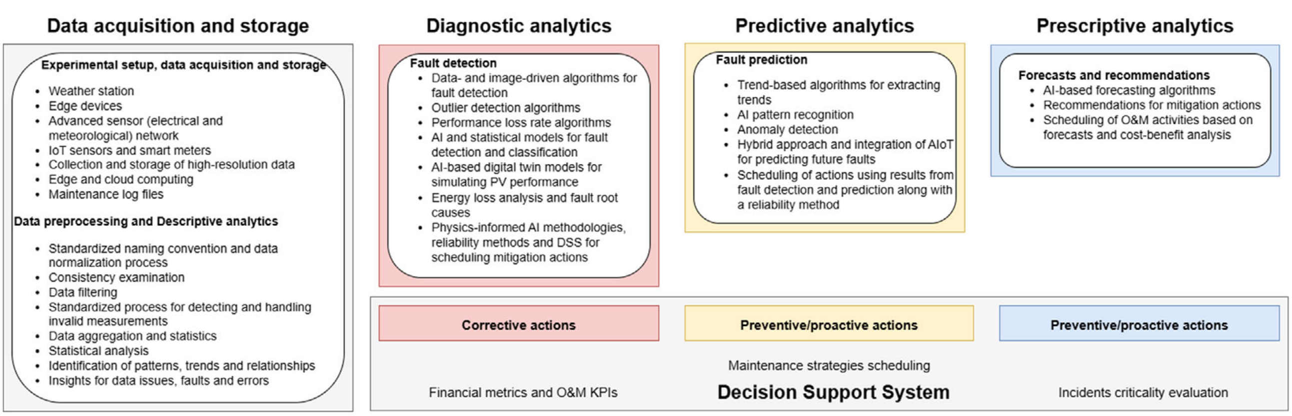

FIGURE 12 | Standardized framework to improve solar asset management.

and robust cybersecurity measures to address new operational challenges introduced by AI adoption [153].

LLMs can be further used for intelligent maintenance approaches in PV systems by integrating autonomous or semiautonomous AI agents that interact dynamically with both data systems and human operators. As AI agents, LLMs can continuously monitor asset performance, interpret real-time sensor data, and cross-reference historical maintenance logs to detect emerging anomalies and recommend context-specific maintenance actions. These agents can proactively generate alerts, schedule interventions, and even initiate procurement workflows for spare parts. By embedding industry-standard KPIs and reliability methods into the reasoning processes, LLM-based agents can prioritize maintenance tasks based on cost-effectiveness and operational risk. Their ability to explain decisions in human-readable language enhances transparency and trust, allowing technicians and asset managers to validate AI-driven recommendations. In this way, LLMs not only automate routine diagnostics and reporting but also act as intelligent collaborators, transforming reactive maintenance workflows into proactive, insight-driven operations. An example of an AI agent approach can be found in [154].

Regarding the new breed of solar-plus-storage systems, dedicated maintenance strategies for battery energy storage systems (BESS) should be developed and integrated into solar asset management frameworks. As energy storage becomes increasingly vital for future energy markets offering smart grid services, its role in facilitating the reliable integration of PV systems into the electrical grid will be indispensable. To ensure long-term performance and operational reliability, future control algorithms and O&M activities should incorporate intelligent analytic capabilities for both PV installations and energy storage systems. This will facilitate the optimal participation of solar-plus-storage systems in grid and market services and will finally enable virtual power plant operation (optimal scheduling and dispatch of units).

4 | Conclusions

Solar PV systems are pivotal in reducing reliance on fossil fuels and promoting a more sustainable energy mix. Their successful integration into the current grid is essential for driving the transition to a greener future. However, despite their widespread adoption, factors such as weather conditions, outdoor exposure, and aging can significantly impair the PV performance over time. Effective O&M actions are thus essential for ensuring long-term stability and for enhancing the PV power output. With the ongoing expansion of the global PV capacity, the demand for intelligent O&M is becoming increasingly critical. The implementation of advanced algorithms will not only minimize the need for human intervention, but it will also significantly streamline O&M procedures, enhancing efficiency and reducing operational costs. By automating key processes, these intelligent algorithms can improve the accuracy of fault detection and prediction, while also optimizing maintenance schedules and field interventions.

Currently, corrective maintenance algorithms, using AI, data- and image-driven fault diagnostic approaches, have been successfully implemented. These approaches utilize the available data/images as inputs and effectively identify different fault types occurring within the PV system. While fault diagnostic and corrective maintenance algorithms are well-established, the development of predictive analytics remains limited due to higher upfront costs compared to corrective maintenance and high number of false alarms.

While there have been only limited efforts in the field to develop predictive maintenance algorithms that generate early warnings and alert technicians before malfunctions escalate, a critical requirement for their successful implementation is the access to high-resolution data and detailed maintenance log files. The literature survey revealed that historical data were the predominant source used in 79% of the reviewed studies, while synthetic and real-time data accounted for 14% and 7%, respectively. This highlights a strong dependence on historical data for the development of predictive maintenance algorithms. Moreover, the main input features to the predictive maintenance algorithms include the PV electrical (e.g., current, voltage, and power) and environmental measurements (such as irradiance, ambient temperature, and module temperature). Apart from the main monitored electrical and meteorological data, other measured/calculated parameters, weather forecasts and maintenance log files could be used as inputs to the predictive models.

A primary challenge in implementing predictive maintenance models is the limited availability of high-quality data and detailed historical maintenance logs. Predictive maintenance relies heavily on accurate, labeled, and extensive datasets to effectively train the models and generate reliable fault predictions. However, many research and industry organizations face difficulties due to insufficient or inconsistent data collection, especially for newly installed assets that lack historical failure records. Data quality issues (such as incomplete logs, sensor inaccuracies, and invalid measurements) further complicate the deployment of predictive maintenance algorithms. As a result, the effectiveness of predictive maintenance initiatives is often constrained by these data limitations, and their accuracy can suffer, leading to a high number of false alarms.

To overcome these challenges, a comprehensive framework for implementing smart maintenance strategies is proposed in this article that provides actionable guidance, recommendations, and future perspectives to improve solar asset management. Following this framework, predictive maintenance could be adopted to enable the early detection and prediction of fault conditions, while also optimally scheduling the field maintenance activities. The successful implementation of predictive maintenance strategies offers substantial financial benefits to the PV plant owners as maintenance expenses will be lowered, while the PV system's energy yield will be improved, making it a profitable investment.

As the deployment of PV systems continues to expand, the integration of intelligent predictive maintenance algorithms for solar-plus-storage systems will become increasingly vital for maintaining grid stability, reducing operational costs, and advancing a sustainable energy transition. Achieving these goals will require effective collaboration among researchers, industry stakeholders, and policymakers to develop standardized frameworks and best practices that facilitate the widespread adoption of predictive maintenance strategies for PV systems.

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Conflicts of Interest

The authors declare no conflicts of interest.

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